For full credit explain your reasoning, showing all relevant work.

Exercise 1. Compute values of the following expressions in the group $U(12)$:

a) $5 \cdot 7$
b) 11^{-1}
c) $5 \cdot 11^{-1}$

Exercise 2. Find all elements of the group $U(20)$ and the inverse for each of them.

Exercise 3. Compute multiplication tables for groups with 4 different elements e, a, b, c (where e is the identity element) in each of the following cases:

1) $a \cdot a = b \cdot b = c \cdot c = e$
2) $a \cdot a = a$,
3) $a \cdot a = b$

Note. All other possible multiplication tables can be obtained from these by relabeling elements. For example, the multiplication table in the case where $a \cdot a = c$ is the same as in the case $a \cdot a = b$, but with b and c swapped.

PRACTICE PROBLEMS

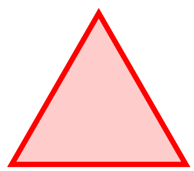
Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Compute values of the following expressions in the group $\mathbb{Z}_{12}$:

a) $10 + 5$
b) $4 + 8$
c) $4 - 8$

Practice Exercise 2. For an integer $n > 2$, show that there are at least two different elements $a, b \in U(n)$ such that $a \odot a = 1$ and $b \odot b = 1$ (where $\odot$ denotes multiplication in $U(n)$).

Practice Exercise 3. **a)** Describe all elements of the group D_3, the dihedral group of symmetries an equilateral triangle.

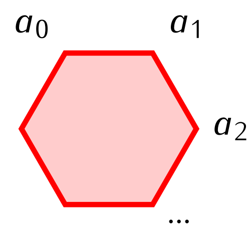


b) Compute the multiplication table of the group D_3.

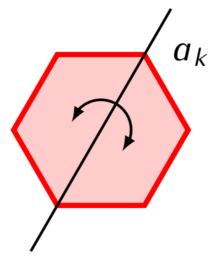
c) Is D_3 an abelian group? Why or why not?

d) For a group G the *order* of an element $g \in G$ is the smallest integer $k > 0$ such that $a^k = e$, where $e \in G$ is the identity elements. Compute the order of each element of D_3.

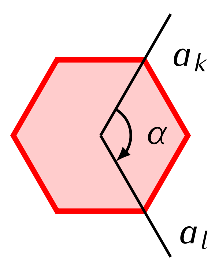
Practice Exercise 4. Let D_n be the dihedral group of symmetries of a regular polygon with n vertices. Label vertices of the polygon by $a_0, a_1, \ldots, a_{n-1}$:



Let $V_k \in D_n$ denote the symmetry that reflects the polygon about the axis passing through the vertex a_k:



Show that the composition $V_l \circ V_k$ is the rotation by the angle 2α where α is the angle from the vertex k to the vertex l:



In other words, $V_l \circ V_k$ is the rotation that takes the vertex a_0 to the vertex a_r where $r = 2(l - k) \mod n$.